PROBLEM CL8

Jet Impinging on a Wall with an Orifice

Original statement · CL8.pdf

Free jets | Bernoulli | Mass | Force on a wall

OFFICIAL PROBLEM STATEMENT

A circular water jet of diameter D_j and velocity V_j impinges perpendicularly to a wall, where there exists an orifice centered at the jet axis, as shown in the figure of CL8.pdf, so that part of the water flows through the orifice and part is deflected to flow next to the wall. Assuming that the flow velocity is sufficiently large for the effects of viscosity and gravity to be negligible:

- Determine the values of the velocities V_s and V_r for the secondary jet that emerges from the orifice and for the radial wall jet.
- Obtain the thickness $h(r)$ of the radial wall jet.
- Compute the force that the jet exerts on the wall as a function of ρ, V_j, D_j and D_s.

Source: CL8.pdf (page 1). Notes.pdf: Bernoulli's equation Eq. 4.17 (Chapter 4), integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible, frictionless (viscosity negligible); neglect $\bar{g}$ (buoyancy-free).
- Free jets surrounded by the ambient air at p_a. In each jet away from the wall the streamlines are straight and parallel (aligned), so the pressure does not vary across them: $p = p_a$ inside the incoming jet, the secondary jet and the wall jet (official solution).
- Data: $[\rho,\ V_j,\ D_j,\ D_s]$. Unknowns: $[V_s,\ V_r,\ h(r),\ \bar{F}_w]$.

Symbol	Meaning	SI unit
V_j, D_j	velocity and diameter of the jet	m/s, m
V_s, D_s	secondary jet velocity, orifice diameter	m/s, m

Symbol	Meaning	SI unit
$V_r, h(r)$	wall-jet velocity and thickness	m/s, m
$\bar{F}_w$	force of the jet on the wall	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the wall; $\bar{e}_x$ along the jet axis (towards the wall); cylindrical coordinate r on the wall, with $\int \bar{e}_r \, d\theta = 0$.
- Fixed control volume around the water: Σ_j (section of the incoming jet, $\bar{n} = -\bar{e}_x$), Σ_s (section of the secondary jet beyond the orifice, $\bar{n} = +\bar{e}_x$), Σ_r (cylinder of radius r across the wall jet, $\bar{n} = \bar{e}_r$, height h), Σ_a (free surfaces, $p = p_a$, $\bar{v} \cdot \bar{n} = 0$) and Σ_w (wetted wall).

Force of the water (and of the ambient air on the dry side) on the wall:

$$\boxed{\bar{F}_w = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

PHASE 3

Bernoulli and Balances, Surface by Surface

3.1 Velocities: Bernoulli along a streamline

Why this tool: for steady, frictionless, incompressible flow without mass forces, $p + \frac{1}{2}\rho|\bar{v}|^2$ is constant along a streamline (Notes Eq. 4.17 with $U = 0$, as used in the official solution). Take a streamline from the incoming jet to the wall jet, and another one to the secondary jet; at both ends $p = p_a$:

$$p_a + \tfrac{1}{2}\rho V_j^2 = p_a + \tfrac{1}{2}\rho V_r^2 \implies \boxed{V_r = V_j}, \qquad p_a + \tfrac{1}{2}\rho V_j^2 = p_a + \tfrac{1}{2}\rho V_s^2 \implies \boxed{V_s = V_j} \quad (1)$$

> Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_j	$-\rho V_j \frac{\pi D_j^2}{4}$
Σ_s	$+\rho V_s \frac{\pi D_s^2}{4}$
Σ_r	$+\rho V_r\, 2\pi r\, h$
$\Sigma_a,\ \Sigma_w$	0

With (1), $V_s = V_r = V_j$ and the velocity cancels.

$$-\frac{\pi D_j^2}{4} + \frac{\pi D_s^2}{4} + 2\pi r h = 0 \implies \boxed{h(r) = \frac{D_j^2 - D_s^2}{8\,r}} \qquad (2)$$

`Linear momentum (Notes Eq. 3.36)`

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_j	$\rho V_j \bar{e}_x(-V_j)\frac{\pi D_j^2}{4} = -\rho V_j^2\frac{\pi D_j^2}{4}\bar{e}_x$	$0\ (p=p_a)$	0
Σ_s	$+\rho V_s^2\frac{\pi D_s^2}{4}\bar{e}_x$	$0\ (p=p_a)$	0
Σ_r	$\rho V_r^2 h\, r\int_0^{2\pi}\bar{e}_r\,d\theta = 0$	$0\ (p=p_a)$	0
Σ_a	0	$0\ (p=p_a)$	0

Surface	$\int \rho \bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}_w$. With (1), $V_s = V_j$.

$$\left(\rho V_j^2\frac{\pi D_s^2}{4} - \rho V_j^2\frac{\pi D_j^2}{4}\right)\bar{e}_x = -\bar{F}_w \implies \boxed{\bar{F}_w = \rho V_j^2\,\frac{\pi}{4}\left(D_j^2 - D_s^2\right)\bar{e}_x} \quad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Only the part of the jet stopped by the wall (the annulus between D_s and D_j) pushes on it; the secondary jet keeps its momentum. For $D_s \to 0$, $\bar{F}_w \to \rho V_j^2 \pi D_j^2/4$, the force of a jet on a flat plate; for $D_s \to D_j$ the jet passes through and $\bar{F}_w \to 0$, $h \to 0$.
- $h \propto 1/r$: the wall jet thins as it spreads, keeping the flow rate $2\pi r h V_j$ constant.
- Dimensions (SI): $[h] = \mathrm{m}^2/\mathrm{m} = \mathrm{m}$; $[\rho V_j^2 D^2] = \mathrm{N}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $V_j = 10$ m/s, $D_j = 0.05$ m, $D_s = 0.02$ m give $|\bar{F}_w| = 164.93$ N and $h(0.1\ \mathrm{m}) = 2.625$ mm.
- Agreement with the official solution: CL8.pdf obtains (1), (2) and (3).